Lucas Neumann

Senior Software Engineer — Pricing, Risk & AI Tooling

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Summary

Senior backend engineer (Python, Go) building data pipelines, cloud infrastructure and AI-agent tooling for financial systems. Designed MCP-based tooling that gives LLM agents safe, read-only access to production data, and owned real-time market-data and pricing services where downtime is not an option.

Experience

BWGI 2025–present

Senior Software Engineer

- Integrated vendor platforms — Arcesium PMS and other third-party data and pricing products — through complex, high-throughput ETL pipelines, with automated reconciliation against legacy pricing.
- Designed a new risk data pipeline — risk-engine output → PostgreSQL → Qlik — with Greeks and risk-factor serving tables and a data-quality gate (cross-layer reconciliation, robust z-score outliers), run daily per fund.
- Built AI-agent tooling for the risk and pricing teams: a Go toolkit of CLIs and **MCP** servers plus 20+ agent skills that give LLM agents governed access to internal data and systems, shipped as versioned releases.
- Led large migrations and refactors across a legacy codebase — orchestration moved to Airflow 3 on Kubernetes, core data stack upgraded across 18 shared packages — while keeping pricing and risk services running in production.
- Main on-call for pricing and risk; about 790 merged PRs across 32 repositories in 11 months.

Bank of America 2023–2025

Software Engineer, AVP — FICC Emerging Markets

Joined via BRQ as consultant in 2023; hired directly as Assistant Vice President in 2024.

- Built and ran real-time curve infrastructure for the EM derivatives and FX trading desk: live market data and trader curve marks feeding the sell-side pricing system, plus intraday and end-of-day curve publication to internal databases.
- Owned the real-time **PTAX** integration with Banco Central do Brasil — a high-availability service with near-zero fault tolerance, where any failure carries direct reputational risk.
- Developed in one of the industry's largest Python monorepos (15+ years, C++ performance layer): event-driven market-data processing on proprietary NoSQL and messaging systems.

Apex Capital 2021–2022

Software Developer

Hedge fund

- Migrated operational routines from Windows Task Scheduler to Apache Airflow, improving reliability and visibility of daily fund processes.
- Integrated financial-service APIs and built dashboards, data visualisations and scrapers for the investment team.

Skills

Engineering
Python, Go, SQL (PostgreSQL, Snowflake), Airflow/ Astronomer, Docker, Kubernetes, AWS, Datadog, MCP, Linux, Git

Financial systems
Bloomberg MARS & market data, Arcesium, MSCI Barra, S&P Global, Qlik

Domain
Rates, credit, FX, fixed income; derivatives pricing; VaR, expected shortfall, Greeks; reconciliation

Languages
Portuguese (native), English (fluent)

Education

UNICAMP — University of Campinas
MSc, Physical Chemistry
2018–2020
Computational simulation of proteins; daily work with HPC clusters, Python, Julia, C/C++ and Linux.

Projects

git-archaeologist (Rust) — Fast CLI extracting time series from git history (LOC, cohort survival, coupling, hotspots) as TSV, CSV, JSON or Parquet.

ragwatcher (Python) — Incremental local RAG over a directory, served as an MCP tool and CLI.

visidata-rs (Rust) — Rewrite of the VisiData terminal data explorer in Rust (Polars, Ratatui), Lua-extensible like Neovim; in progress.

StringArt.jl (Julia) — Generative art: turns photos into string-art patterns.

MonteCarloArt.jl (Julia) — Generative art: pointillist images via Monte Carlo sampling.

belousov-zhabotinsky (Python) — ASCII-art simulation of the Belousov–Zhabotinsky oscillating reaction.